

Advanced Theory — Module 08: Planning — Study Questions

1. List the five timescales of free energy minimization (perception, action, attention, learning, development). What variable is optimized at each?
2. How does PID control arise as a special case of Active Inference?
3. How does expected free energy $G(\pi)$ generalize the cost function in optimal control?
4. What does Active Inference add beyond model predictive control?
5. Compare reinforcement learning and Active Inference on: reward, exploration, exploitation, and model use.
6. How does epistemic value provide built-in exploration without ϵ -greedy or curiosity bonuses?
7. What is the formal analogy between Helmholtz free energy and variational free energy?
8. How does temperature in thermodynamics correspond to precision in Active Inference?
9. What is the Active Inference analogue of the second law of thermodynamics?
10. How does mutual information relate to epistemic value in expected free energy?
11. What is rate-distortion theory? How does the complexity cost in free energy correspond to the "rate"?
12. What is Prigogine's minimum entropy production principle? How does it connect to Active Inference at NESS?
13. What necessary conditions for consciousness does Active Inference provide? Why are they insufficient?
14. What scalability challenges does Active Inference face for real-brain-scale implementation?
15. How can creativity and novelty emerge from a gradient descent framework?
16. What would falsify Active Inference? Why is this question important and difficult?
17. Can Active Inference serve as a blueprint for AGI? What additional elements are needed?
18. How do the mathematical tools from this track (variational calculus, information geometry, Bayesian mechanics, RG, BMS, multi-agent) interconnect?
19. What is the relationship between Active Inference and predictive processing? Are they the same theory?

20. Write a synthesis: What is the explanatory scope of Active Inference, and where are its genuine limits?